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GitHub

By Akhil Khandelwal

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Last lecture, Git gave us a time machine for our code. Where does that repository live right now?

Where git lives

Your entire history, every commit, every snapshot is inside a **hidden folder (.git)** on one physical machine - your Laptop.

Git Repository

(Commits, history everything)

Your Laptop

(All your history. All your work)

What if this laptop is damaged or lost?

What if this laptop is damaged or lost?

Git Repository

(Commits, history everything)

Everything is gone

No backup. No copies. No recovery.

The Collaboration Problem

Both developers have made changes into **different files**. How do they combine them into a single project?

Akash's Laptop

Project A - login.js ✏️ modified

Priya's Laptop

Project A - payment.js ✏️ modified

The Collaboration Problem

Both developers have made changes into **different files**. How do they combine them into a single project?

They need to **transfer their changes** to each other somehow.

Using:

- 📧 Email
- WhatsApp
- 💾 USB Drive
- ☁️ Google Drive

The Collaboration Problem

What if both edited the **same file**. How do we combine their work? Whose version wins?

Akash’s Laptop

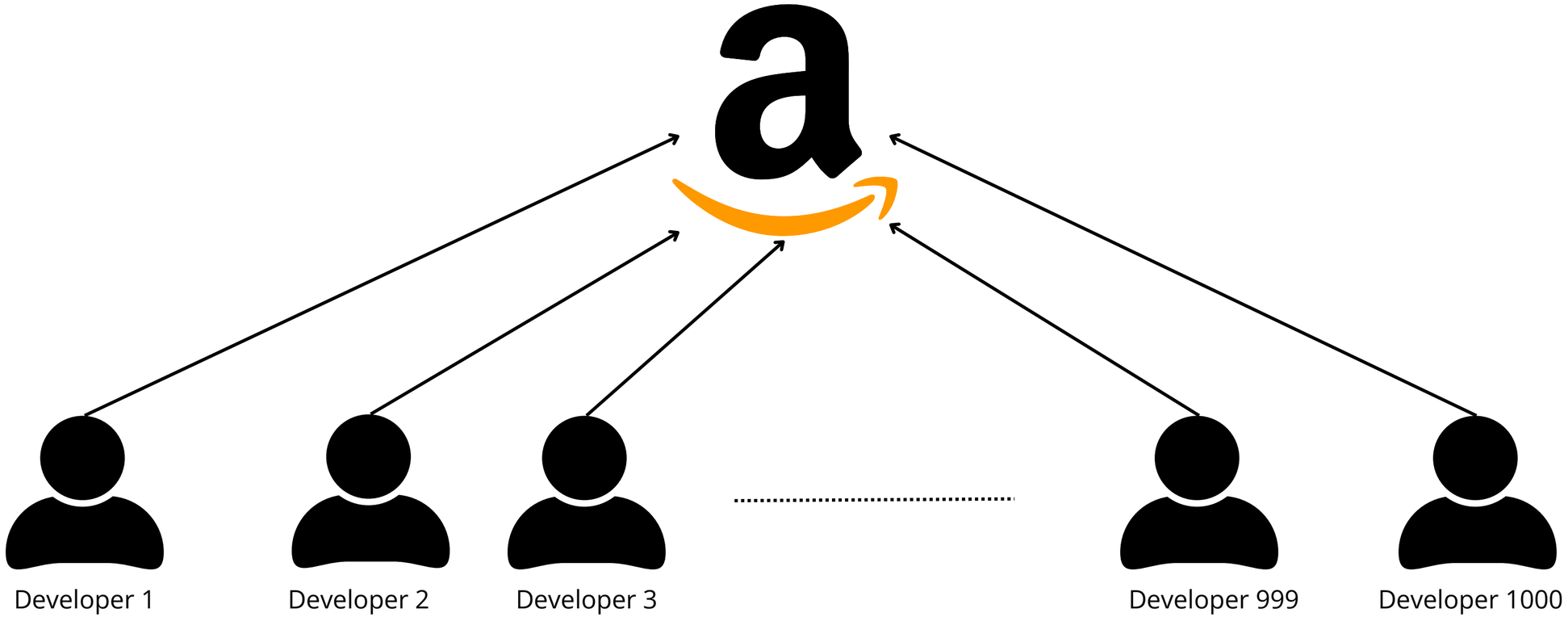
Project A - login.js ✏️ modified

Priya’s Laptop

Project A - login.js ✏️ modified

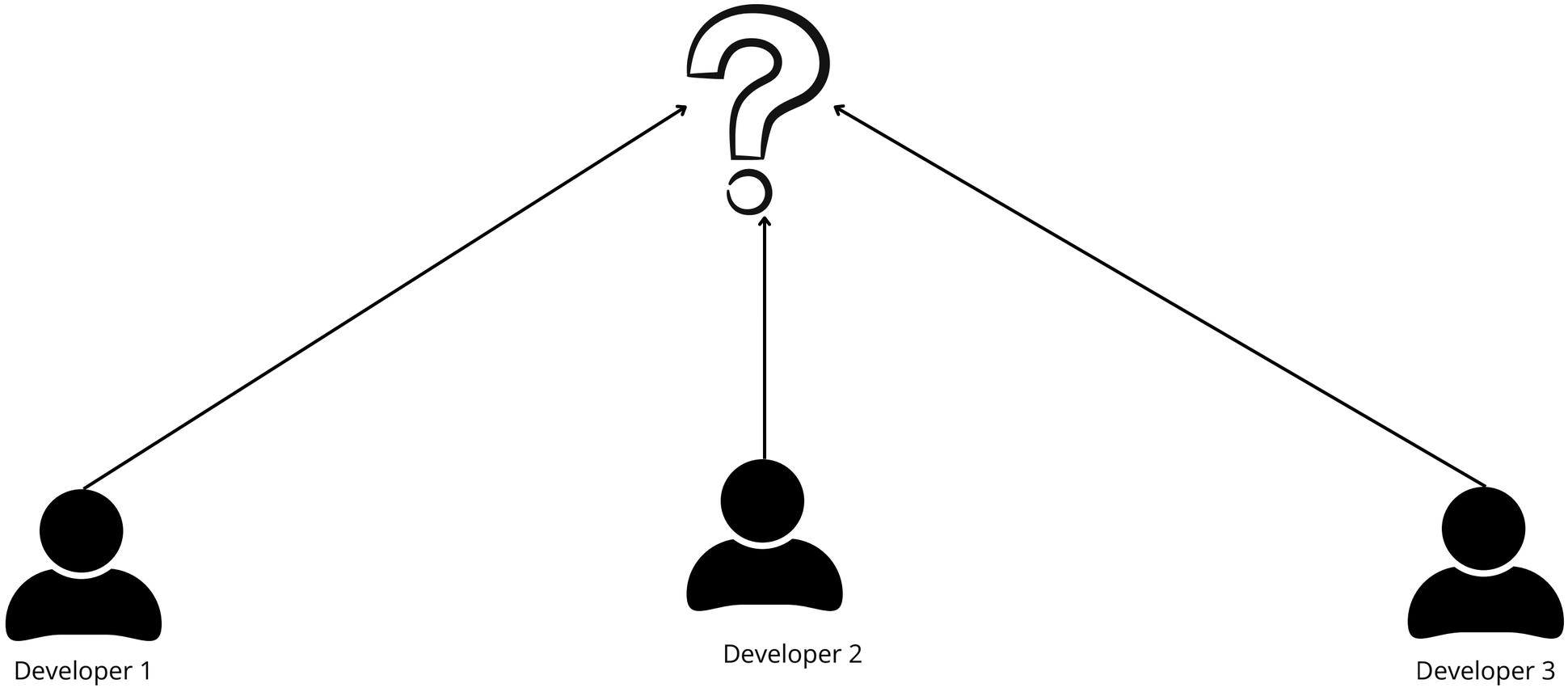
Every App Is Built by a Team

Amazon has hundreds of engineers. Teams of engineers all over the world are editing its code simultaneously. How does that not collapse into chaos?



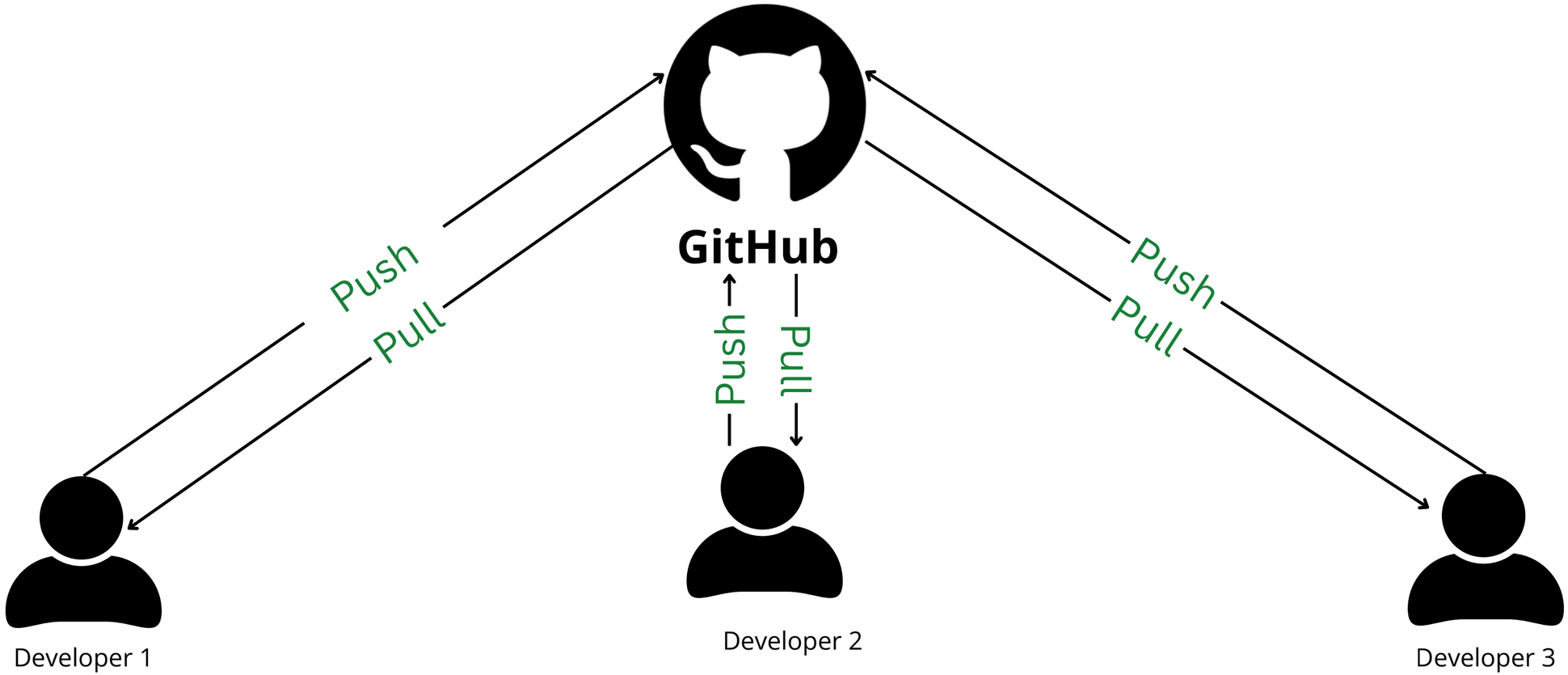
What sits in the middle?

Multiple developers work on the **same project**. How do they **stay in sync**?



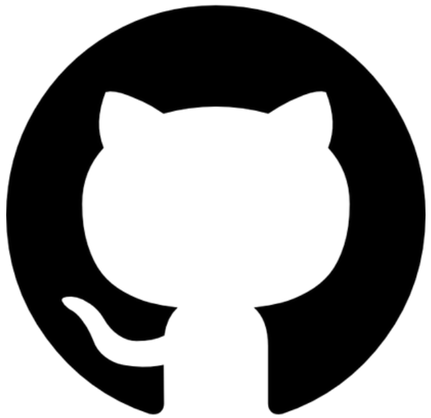
The Missing Piece: A Remote Repository

A central place where developers store and share their code - **Github**.



What is GitHub?

- Most popular platform for **hosting remote Git repositories**.
- Used by developers to store, share, and collaborate on code.
- Hosts the source code for popular open-source projects like **VS Code, React, Python, and Linux**.



Git ≠ GitHub

- **Git** → Version Control Tool
- **GitHub** → Platform for hosting Git Repositories

Git vs GitHub

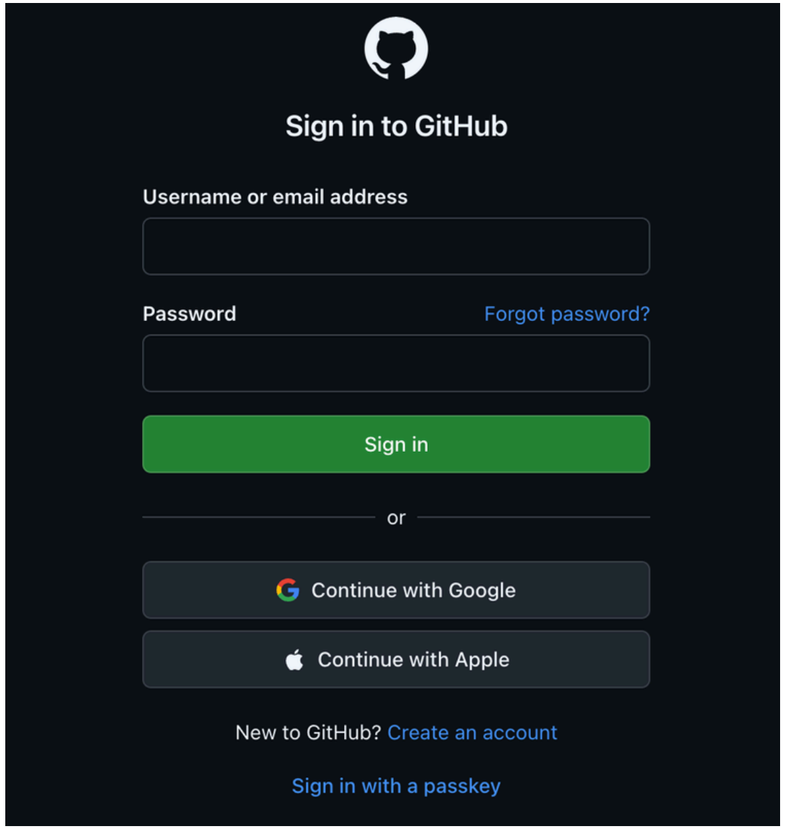
Aspect	Git	Github
Purpose	Version Control for tracking changes	Hosting, collaboration and sharing
Location	Local (On your computer)	Cloud based, Accessible online
Collaboration	Basic collaboration through local git setups	Advanced collaboration tools (pull requests, etc.)
Access	Command Line or GUI	Web Interface with Git Integration
Offline Access	Fully Offline Capability	Requires internet for folder access

Create a Repository on GitHub

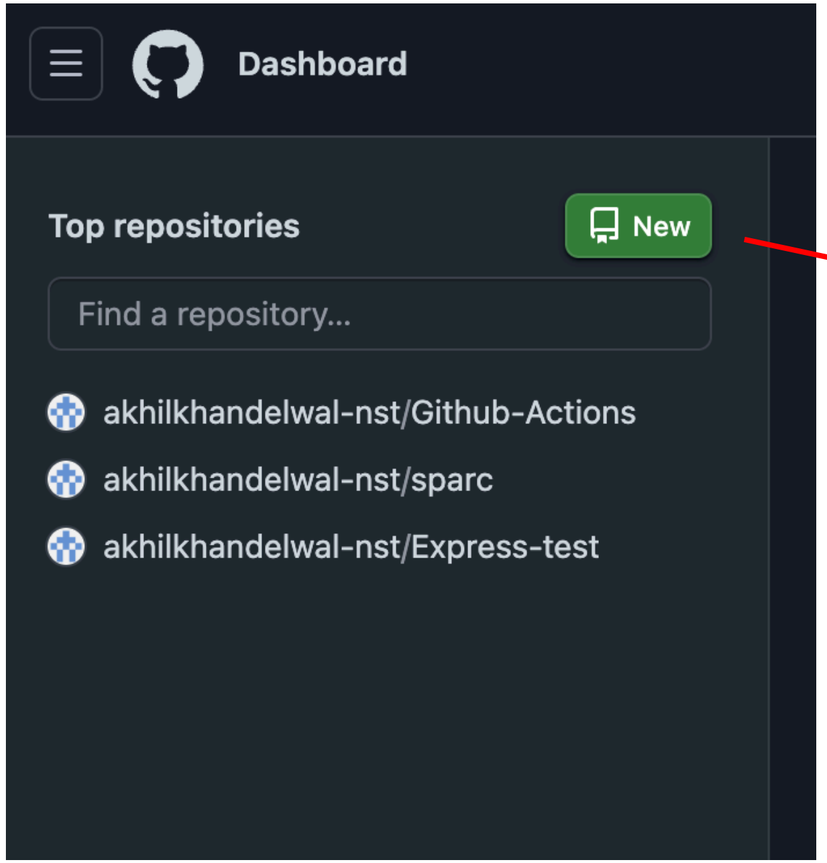
First, we need to create or set up a remote repository on **GitHub** where we can add our changes.

Follow these steps to setup a **remote GitHub repository**.

Step 1: Go to GitHub Website. Enter your credentials and sign in.

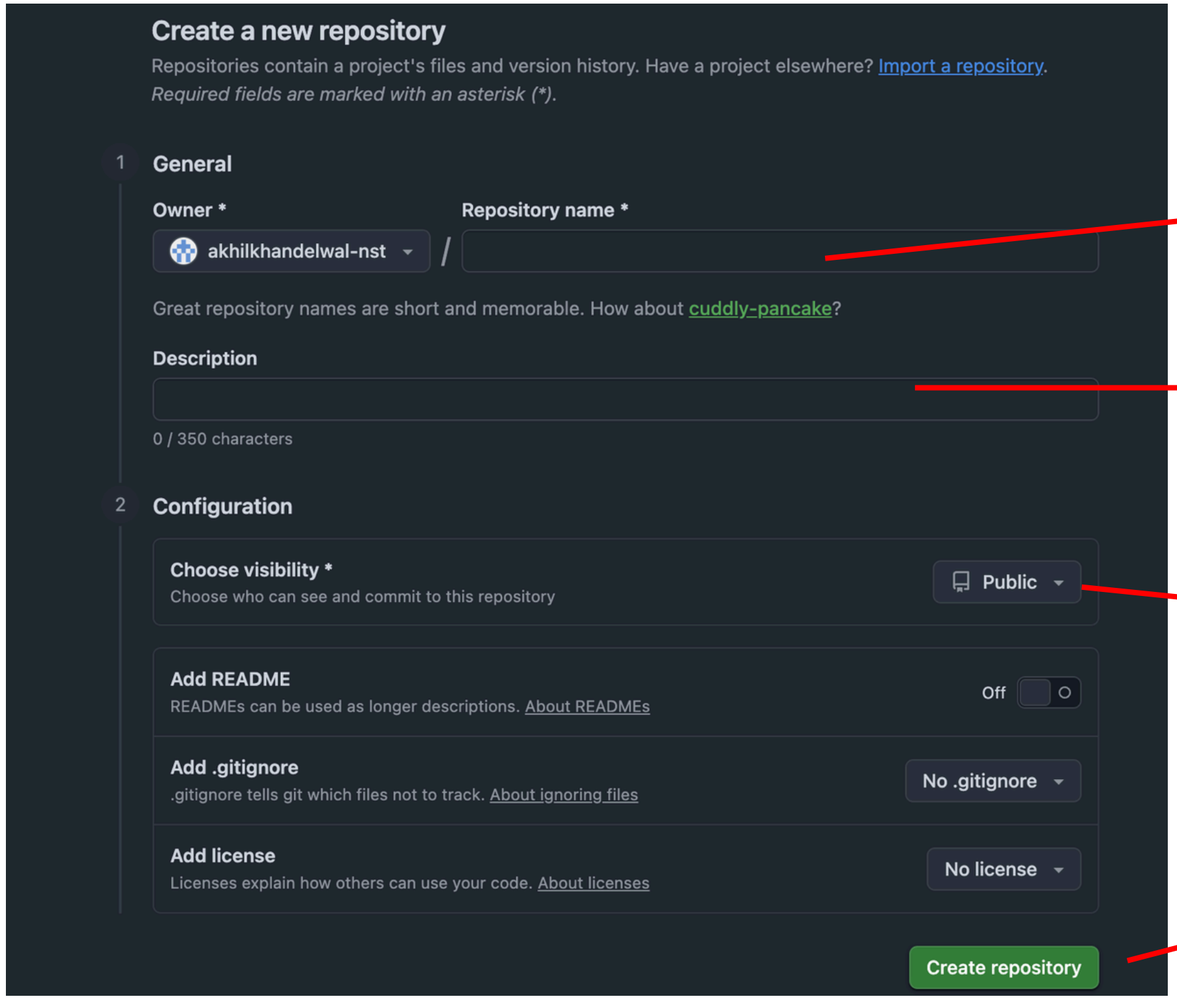


Create a Repository on GitHub



Step 2: Click on “New” to create a new github repo.

Create a Repository on GitHub



Step 3: Enter the details.

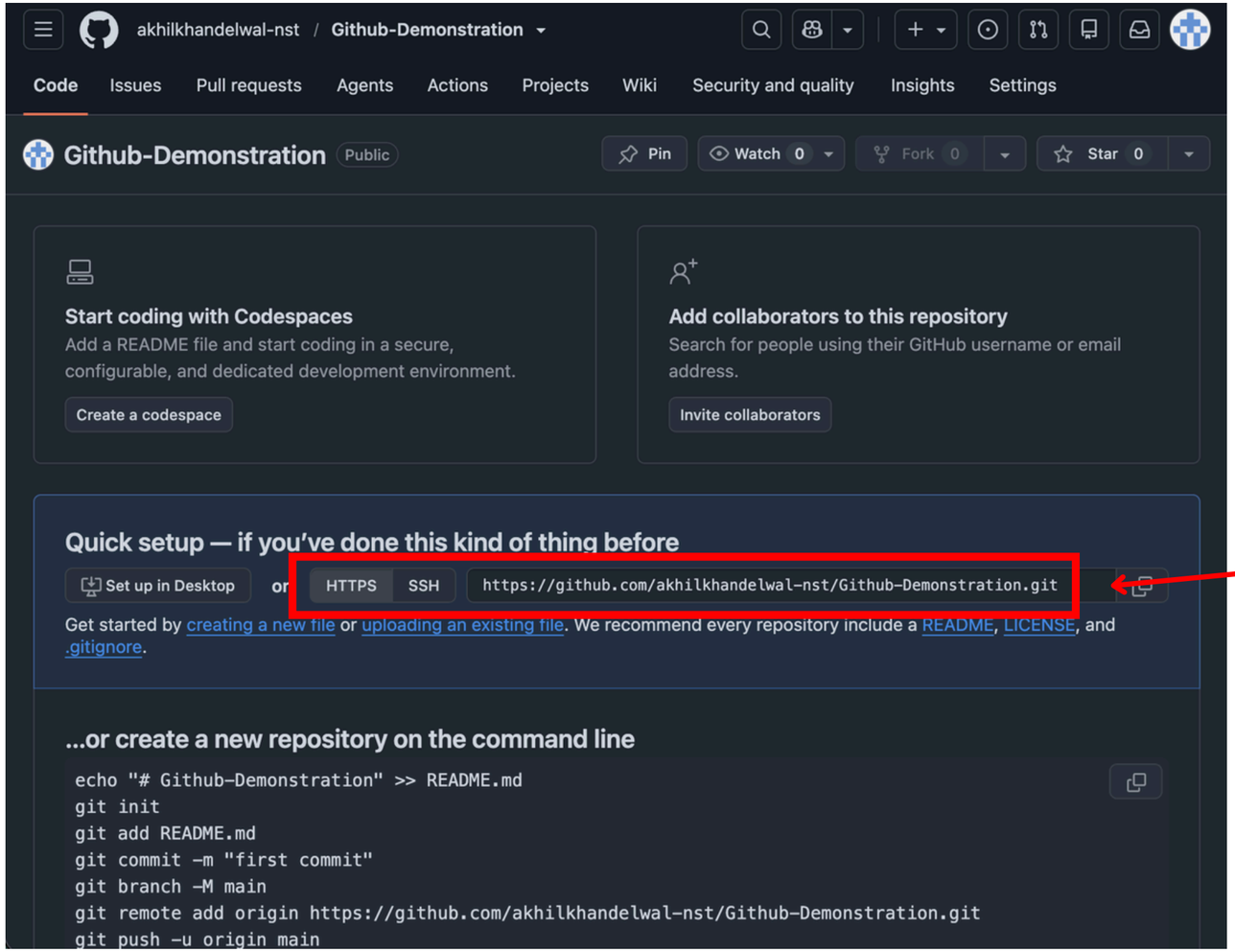
Enter the **name** of repository

Enter the **description** (Optional)

Select visibility: Public = anyone can see it. Private = only you (and people you invite)

Click on “**Create Repository**”

Create a Repository on GitHub



Step 4: Your repo home page would open up.

This is the **address of the remote.** You will give this address to Git so it knows where to send your commits.

Link the Local Repo to GitHub

We need to tell Git where the remote is. We do that with **git remote add.**

```
git remote add origin https://github.com/akhilkhandelwal-nst/Github-Demonstration.git
```

The command to register a new remote address.

The local nickname (alias) you give to that remote.

the address of the GitHub repo

Link the Local Repo to GitHub

Display all configured remote repositories and their fetch/push URLs for your local Git repository.

```
akhilkhandelwal@Akhils-MacBook-Pro-2 my-project % git remote -v
origin  https://github.com/akhilkhandelwal-nst/Github-Demonstration.git (fetch)
origin  https://github.com/akhilkhandelwal-nst/Github-Demonstration.git (push)
```

- **fetch** → Download changes from GitHub
- **push** → Upload changes to GitHub

git push — Send Your History to GitHub

Upload the local main branch to the remote repository named origin and remember this connection for future pushes.

```
git push -u origin main
```

- **git push:** Push local changes to GitHub.
- **-u:** Set the upstream (Remember this remote & branch)
- **origin:** The local nickname (alias) you give to remote.
- **main:** The branch name

git push — Send Your History to GitHub

First time: git push -u origin main

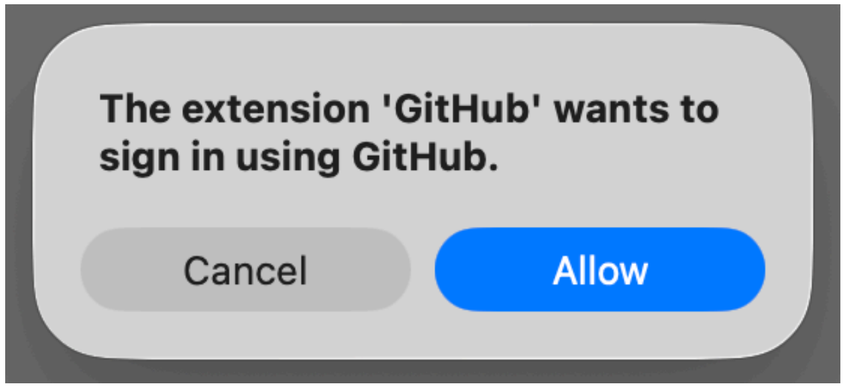
Future pushes: git push

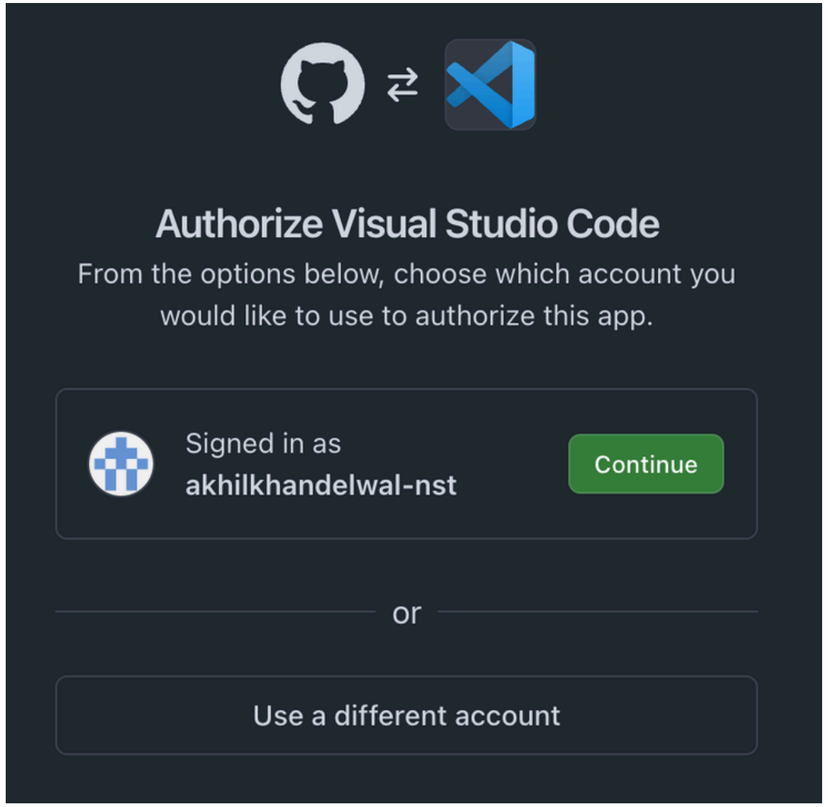
Why does this work?

- -u created a tracking relationship.
- Git remembers:
  - **Remote** → origin
  - **Branch** → main
- Future pushes automatically use origin/main.

GitHub Authentication Process

Step 1: When you push code for the first time, Git asks for **authentication.** You will see this popup in VS Code.

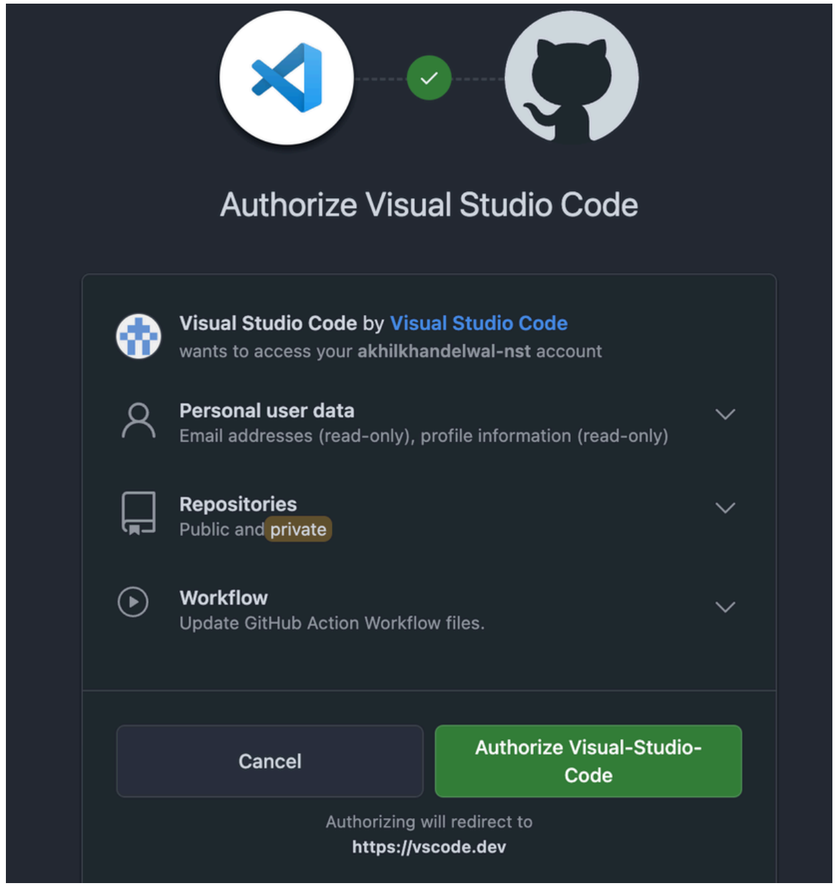




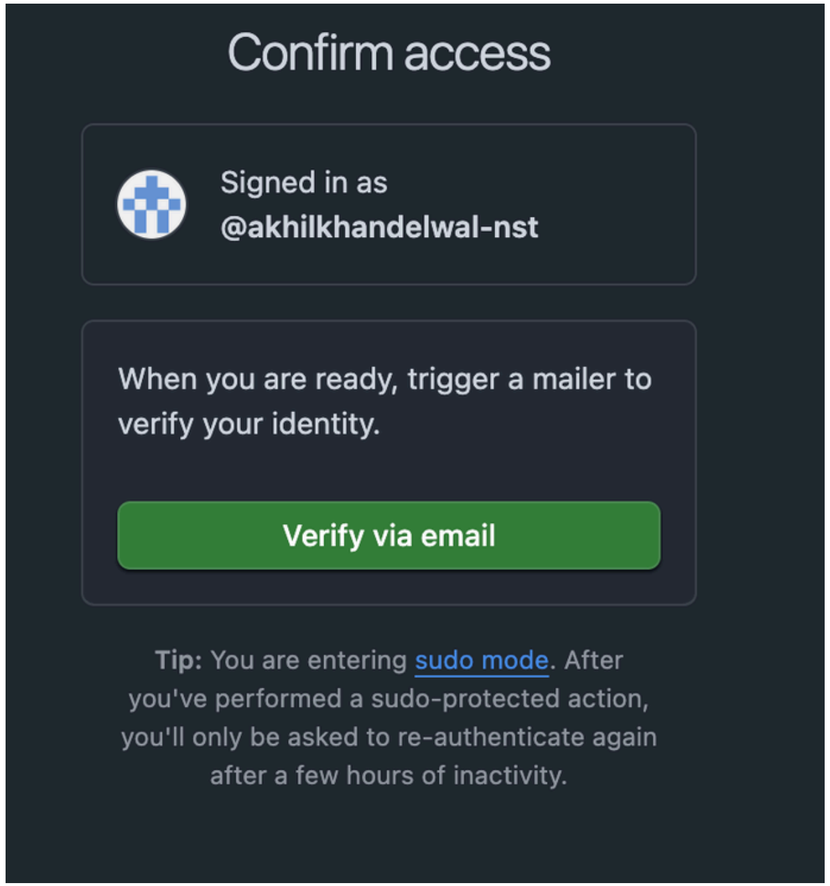
Step 2: A browser window opens for **GitHub login**

GitHub Authentication Process

Step 3: Authorize VS Code

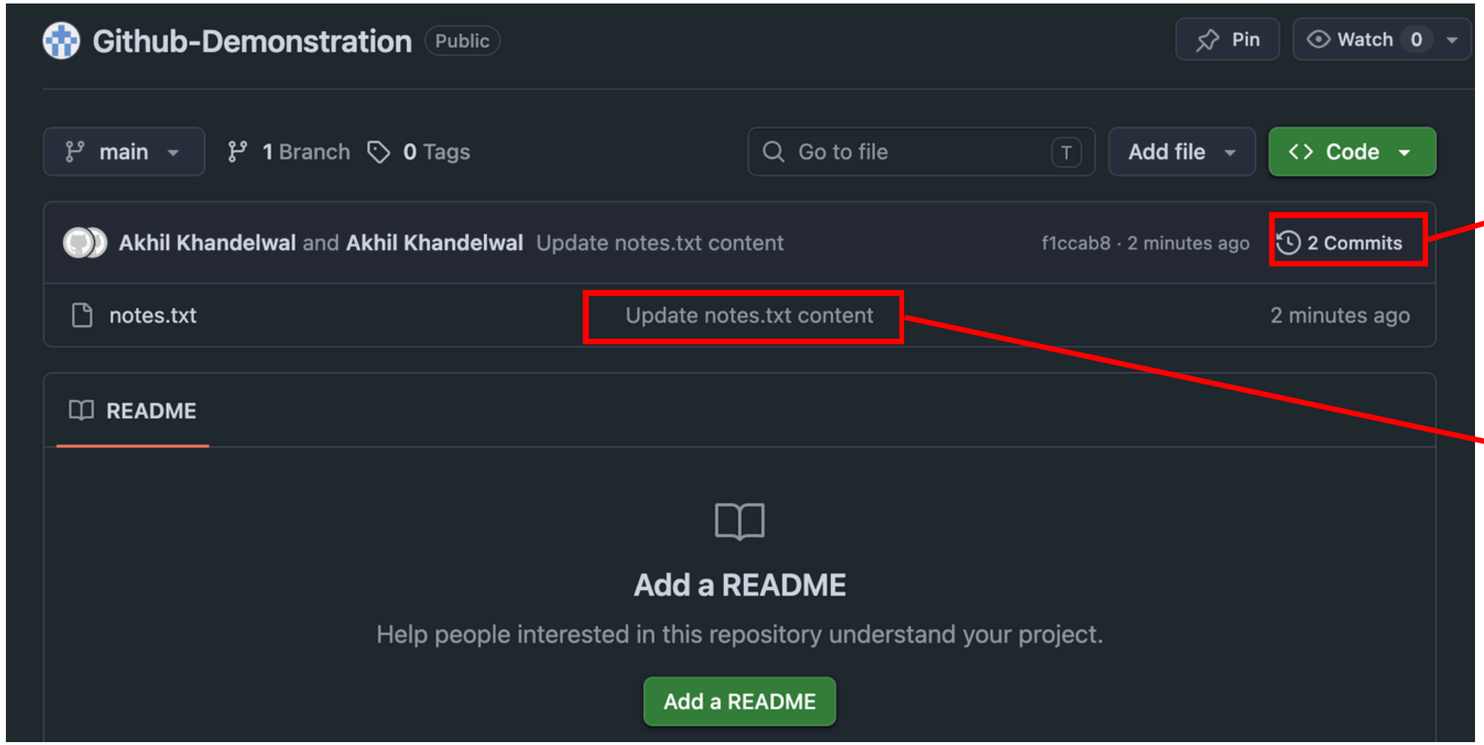


Step 4: Confirm Access



Your code is now pushed to GitHub

Commits are live. Your history is now in two places. **Your machine and GitHub**. If your laptop dies tomorrow, your work survives.

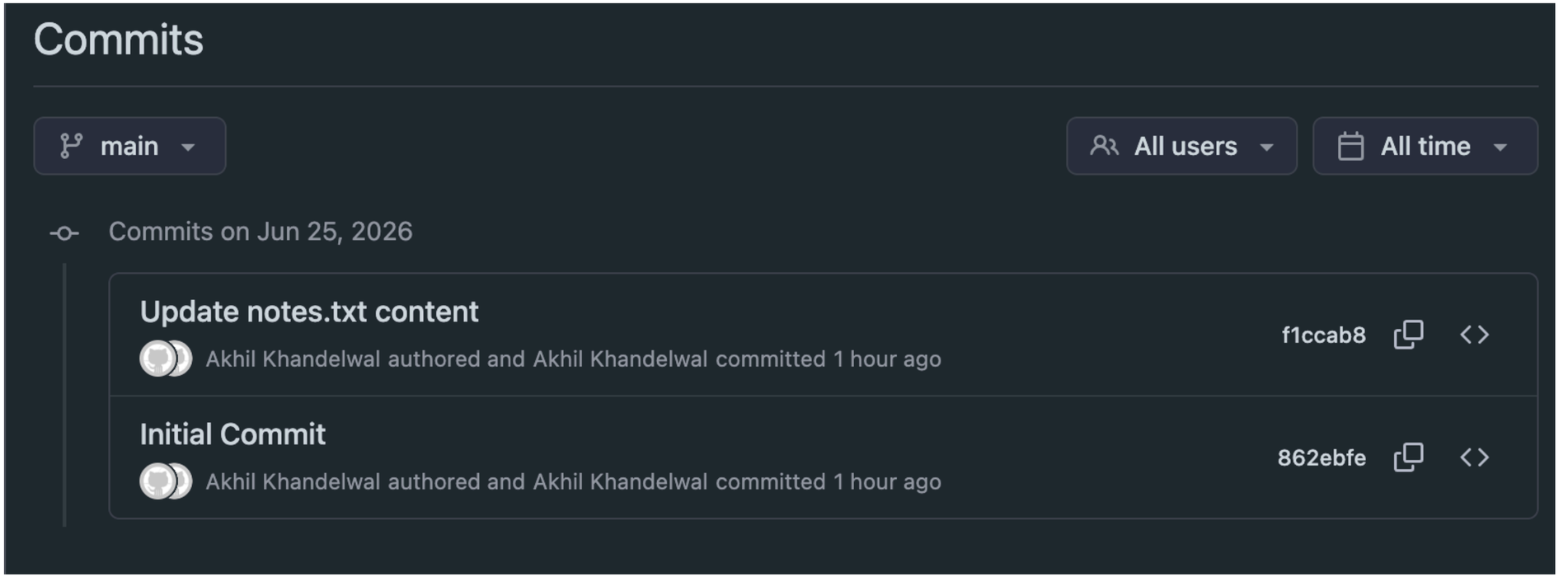


Total commits on the Repository

Latest commit message for notes.txt file.

Commit History on GitHub

- GitHub stores the complete **history of changes.**
- You can view **author**, **time**, and **commit message**.



Why Commit Messages Matter

- Commit messages explain what and **why a change was made**
- Helps teammates understand changes without reading code
- Helps to **remember the context**
- Good messages help trace bugs quickly

Bad Commit Message:

updated file

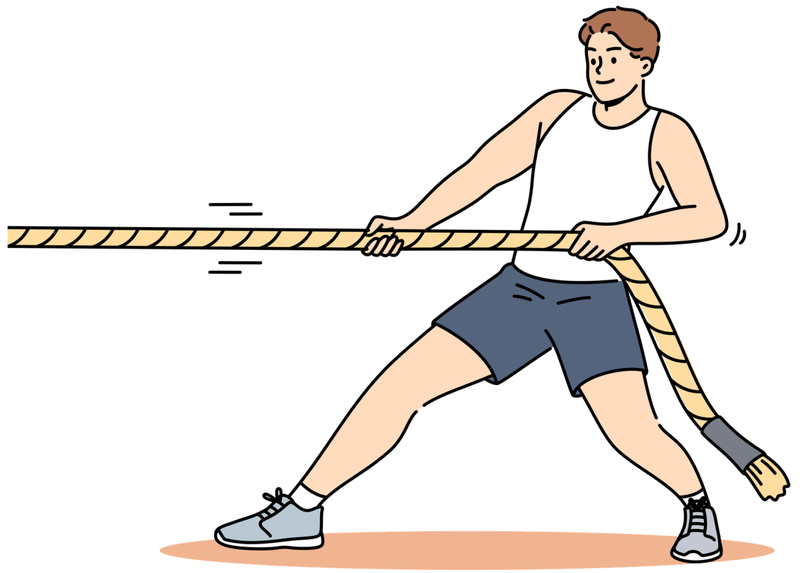
Good Commit Message:

Add student registration form validation

git pull — Getting Changes from GitHub

- **Downloads latest changes** from remote repository (GitHub)
- Updates your l**ocal code**

```
git pull
```

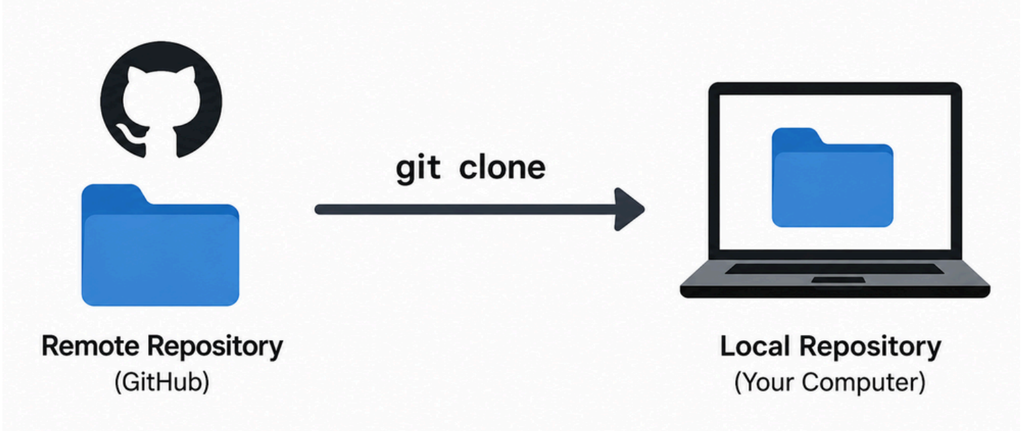


Example: Suppose your teammate adds a login feature and pushes it to GitHub. Before you start working, run git pull to get their latest code on your computer.

git clone — Getting a Repo from GitHub

What does git clone do?

- Downloads a repository from GitHub.
- Creates a local copy on your machine.
- Includes **all files, branches, and commit history.**




```
git clone https://github.com/lazyjinchuriki/pokedex.git
```

Clone = Download the project + its complete Git history

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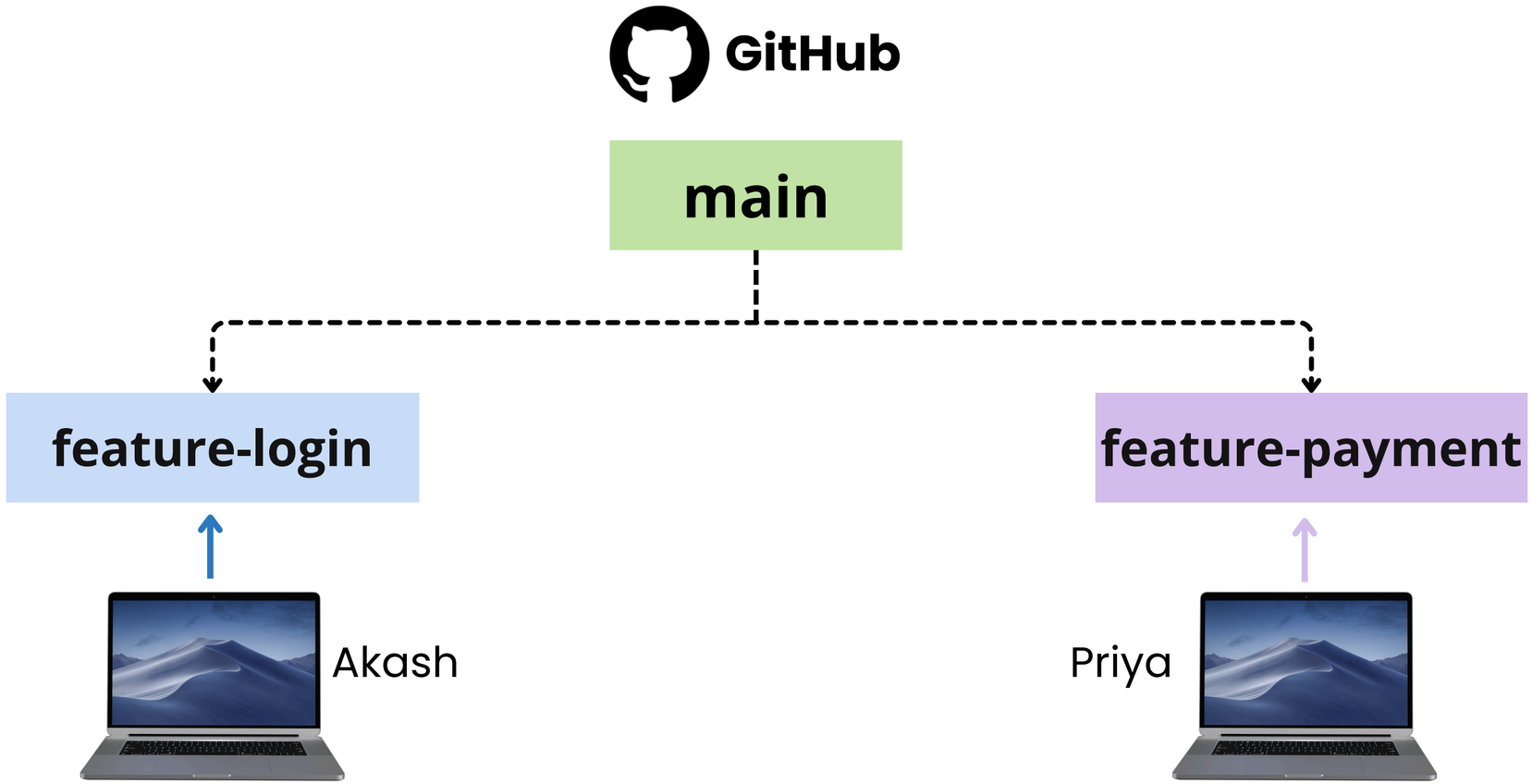
Quiz

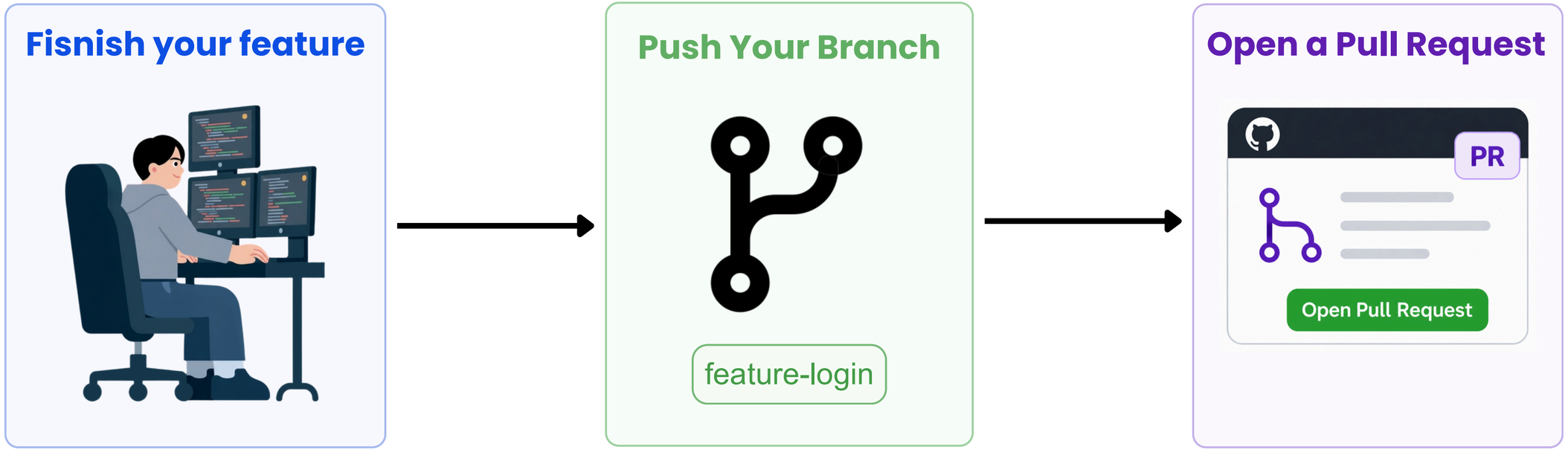
How Teams Use GitHub

Each developer works on **their own feature branch**, not directly on main.



Open a Pull Request

When a developer finishes a feature, they push their branch and open a **Pull Request** for review.



Develop your feature and test it locally.

Push your code to your feature branch.

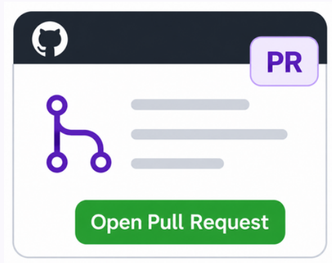
Open a Pull Request to merge into main.

Code Review Before Merge

A teammate reviews the **Pull Request,** leaves feedback, and approves the changes.

Then we merge into main.

Open a Pull Request



Open a Pull Request to merge into main.

Review Changes

Teammates review your changes and provide feedback.

Merge into main

Once approved, your changes are merged into main.

Pull Latest Changes

Everyone can pull the latest changes.

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Live Demonstration: A Real GitHub Project

Live Demonstration: A Real GitHub Project

We'll explore a real **open-source repository** and identify:

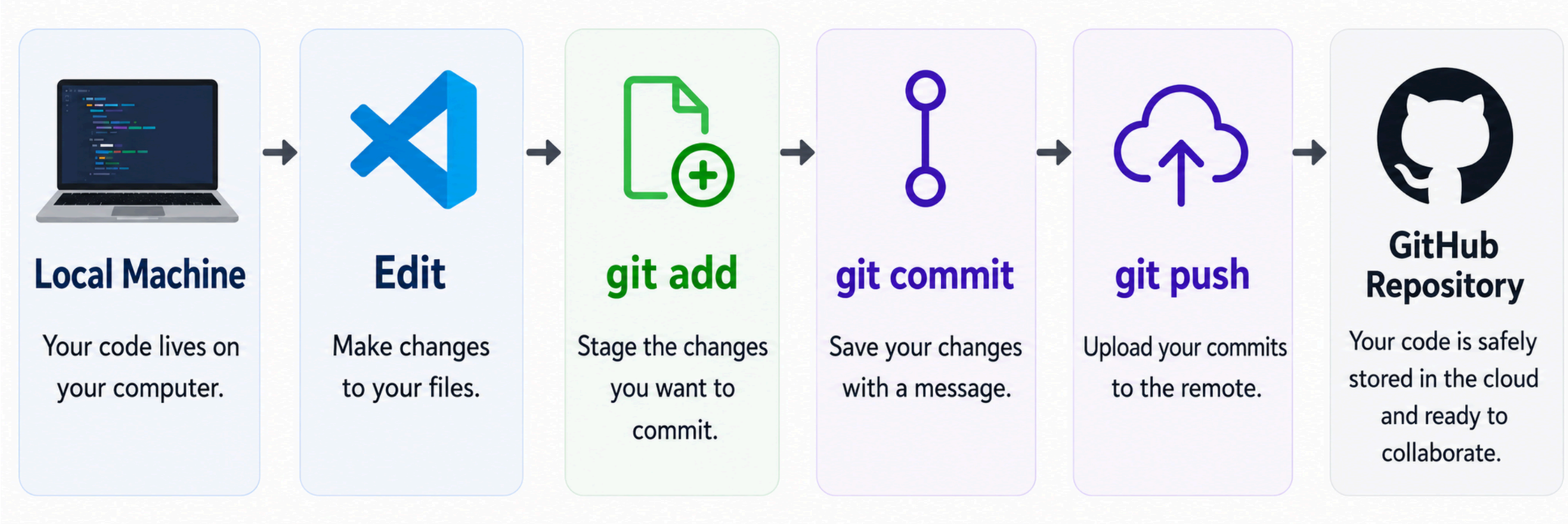
- Commit history
- Contributors
- Pull Requests
- Code Reviews

🔗 github.com/torvalds/linux

Same workflow. Different scale.

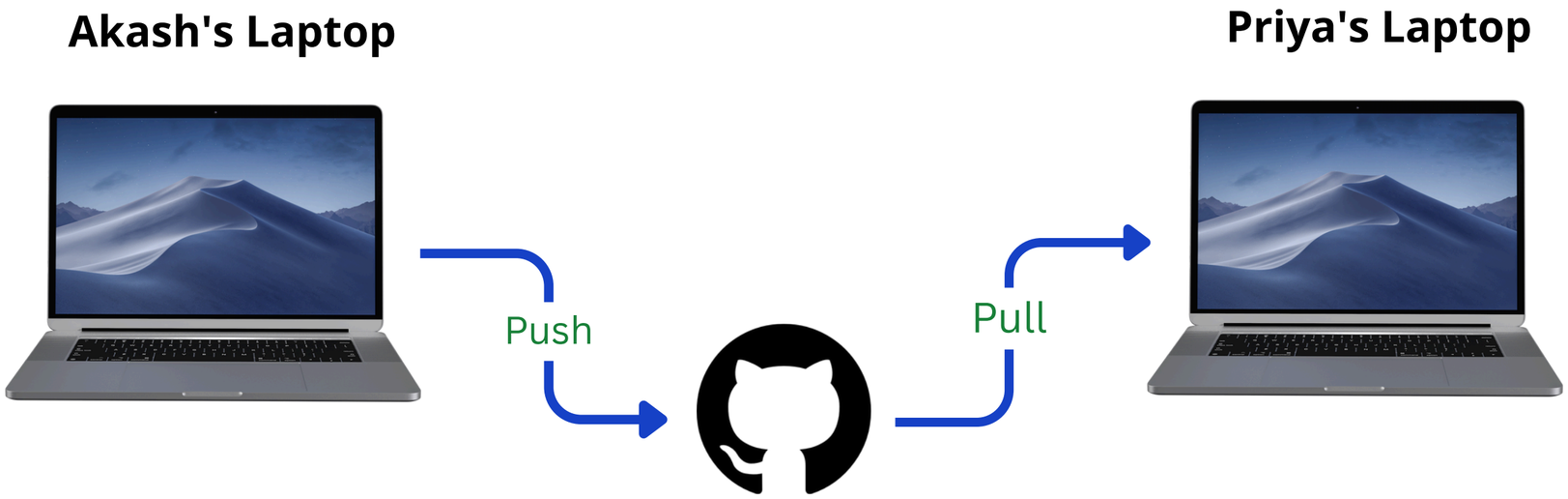
The Full Picture: From Local to Global

Your code starts on your laptop and moves to **GitHub**.



This workflow helps you keep your code **safe**, **organized**, and **easy to collaborate**.

The Full Picture: GitHub to Team



- **GitHub** keeps the team in sync.
- One developer **pushes** changes. Other developers **pull** the latest code.
- Everyone works from the same project repository.

Where this came from ?

GitHub started with a simple idea and grew into the world's most widely used platform for developers.

2008

Github Founded

Founded by 4 developers to make Git easier and more powerful for teams.

2018

Microsoft Acquired GitHub

A major milestone that accelerated GitHub's growth and impact.

Today

100M + Repositories

Millions of developers.
Millions of ideas.
One platform.

Built on GitHub

GitHub started with a simple idea and grew into the world’s most widely used platform for developers.

Linux Kernel

The heart of open source. Built and improved by thousands of contributors worldwide.

Python

A powerful programming language used by millions, built and maintained by a global community.

React

A JavaScript library for building modern user interfaces, used by millions of developers.

VS Code

A lightweight but powerful code editor loved by developers around the world.

Further Reading

1. https://git-scm.com/book/en/v2
2. https://skills.github.com/
3. https://ohmyz.sh/
4. https://github.com/dahlbyk/posh-git
5. https://git-scm.com/
6. https://github.com/

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Thanks for watching!

Feedback

Now it's time to give your valuable feedback